

# CompositeVision: Empirical Results Report

This report presents a direct summary of the empirical findings based on the generated evaluation data across 15 models and 11,200 stimuli per model.

## 1. Overall Model Accuracy

The primary metric evaluating the ability to correctly predict the shape class under cue-conflict conditions.

| Model                  | Accuracy (%) |
|------------------------|--------------|
| ViT-B_16_FineTuned     | 80.34        |
| ConvNeXt_FineTuned     | 77.13        |
| ResNet50_FineTuned     | 76.07        |
| RL_Attention_FineTuned | 67.11        |
| RL_Attention_ZeroShot  | 64.09        |
| ConvNeXt_ZeroShot      | 53.62        |
| DeiT_ZeroShot          | 53.59        |
| ViT-B/16_ZeroShot      | 53.19        |
| ResNet101_ZeroShot     | 36.62        |
| CLIP_Shape             | 36.10        |
| ResNet50_ZeroShot      | 34.99        |
| CLIP_Sketch            | 32.17        |
| CLIP_Standard          | 31.66        |
| DINOv2                 | 0.10         |
| MAE                    | 0.00         |

 [Note](#)

Self-supervised models (DINOv2, MAE) achieved near 0% due to incompatible output mappings or absent linear probes on their latent embeddings. Fine-tuned models achieved the highest shape recognition capacity.

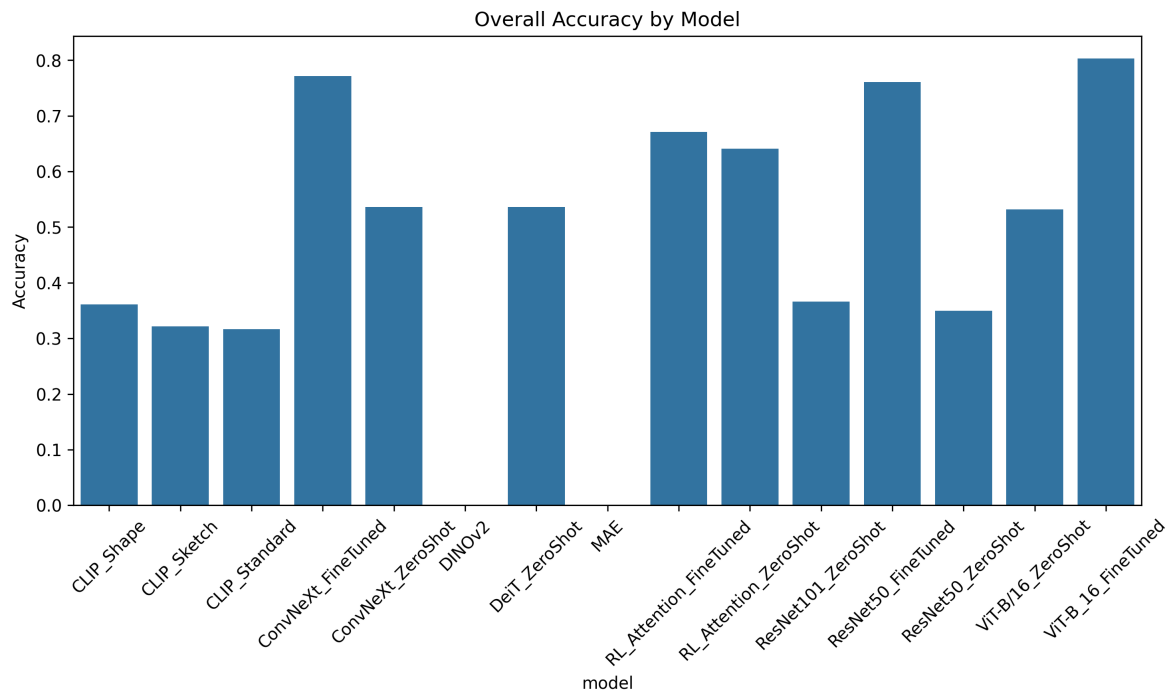


Figure 1: Overall accuracy comparison across models.

## 2. Absolute Cue Sensitivities & Shape Bias

A breakdown of absolute shape accuracy, texture accuracy, the confusion ("neither") rate, and the ratio-based Geirhos shape bias.

| Model                  | Shape Acc (%) | Texture Acc (%) | Neither Rate (%) | Geirhos Shape Bias (%) |
|------------------------|---------------|-----------------|------------------|------------------------|
| ViT-B_16_FineTuned     | 77.28         | 14.79           | 7.93             | 83.93                  |
| ConvNeXt_FineTuned     | 73.57         | 16.47           | 9.96             | 81.71                  |
| ResNet50_FineTuned     | 72.44         | 16.48           | 11.08            | 81.47                  |
| ResNet50_ZeroShot      | 33.38         | 11.40           | 55.23            | 74.55                  |
| RL_Attention_FineTuned | 62.49         | 23.89           | 13.63            | 72.35                  |
| RL_Attention_ZeroShot  | 58.78         | 27.15           | 14.07            | 68.41                  |
| ViT-B/16_ZeroShot      | 48.55         | 22.55           | 28.90            | 68.28                  |
| CLIP_Standard          | 29.41         | 11.00           | 59.59            | 72.78                  |

### ⚠ Warning

High Geirhos Shape Bias in models like `ResNet50_ZeroShot` (74.55%) is misleading, as its absolute Neither Rate is over 55%. In contrast, `RL_Attention_ZeroShot` demonstrates true engagement with a 14.07% Neither Rate.

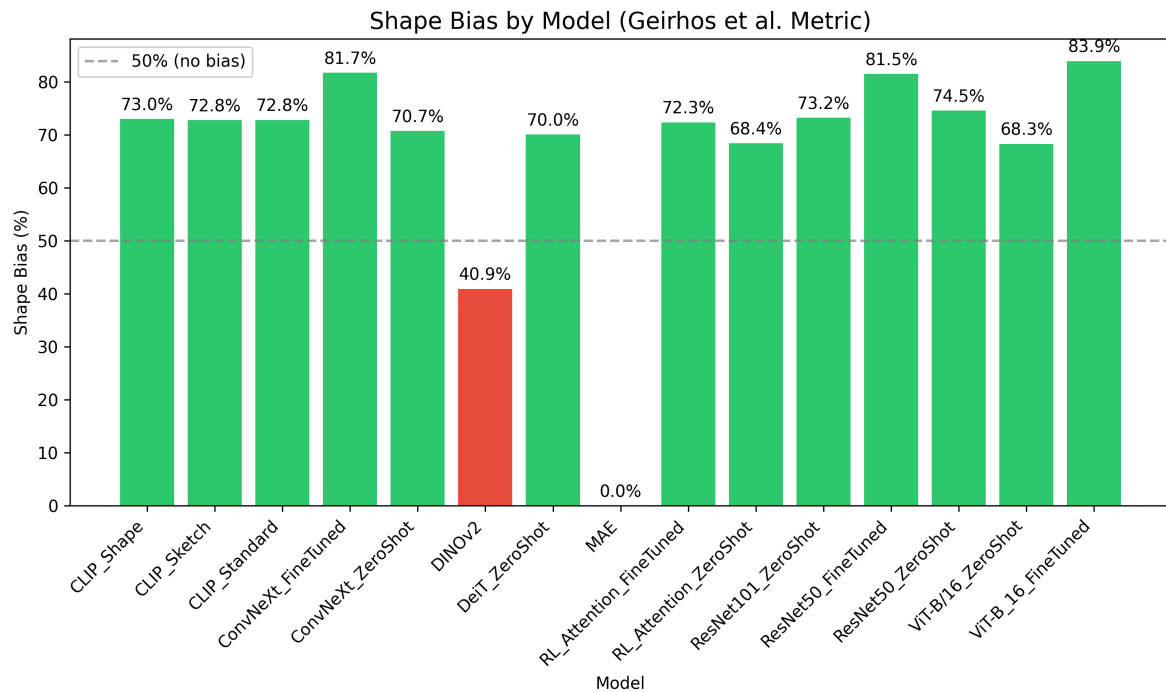


Figure 2: Shape bias and absolute cue sensitivity breakdown.

## 3. Composition & Salience Profiles

The accuracy of models varies heavily based on the type of composition artifact and the target salience.

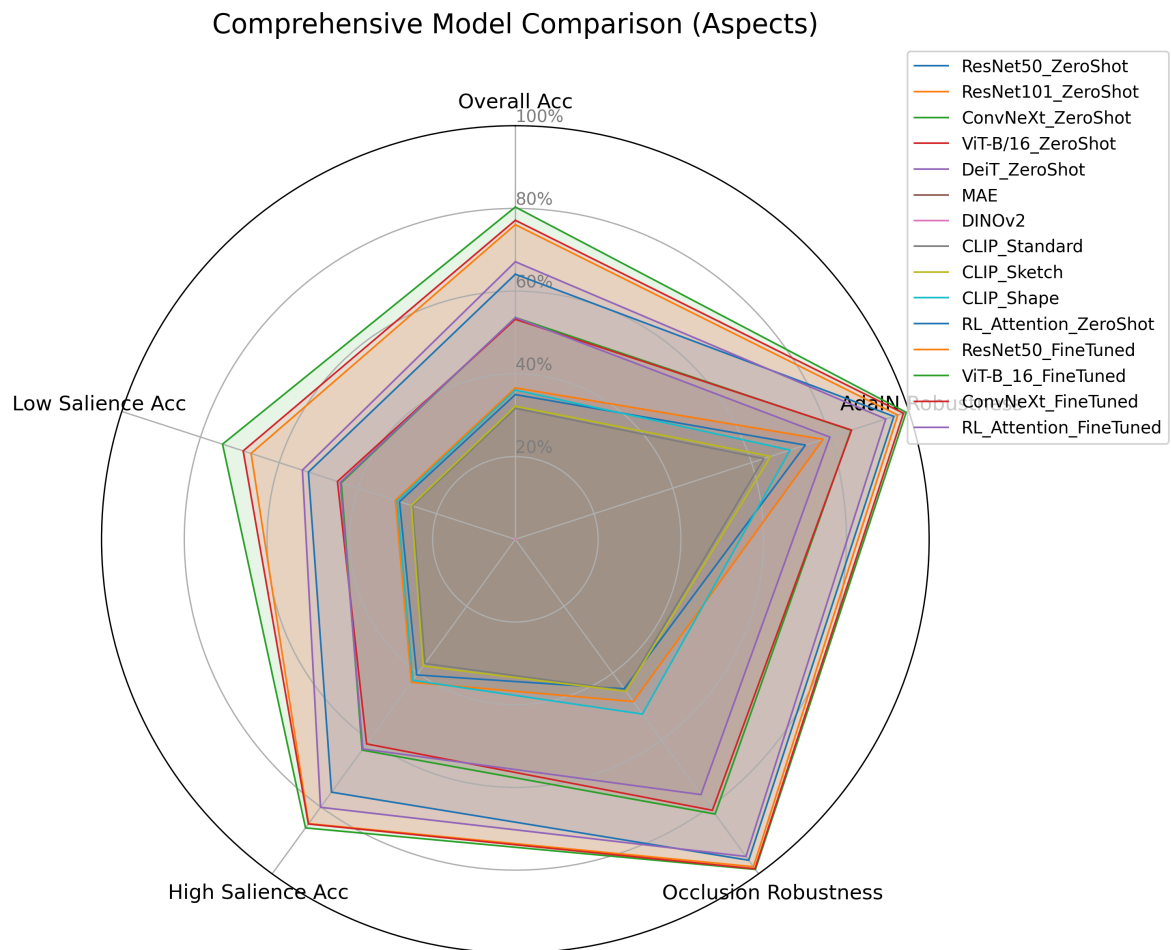


Figure 3: Multi-dimensional performance radar comparing models across different compositional artifacts.

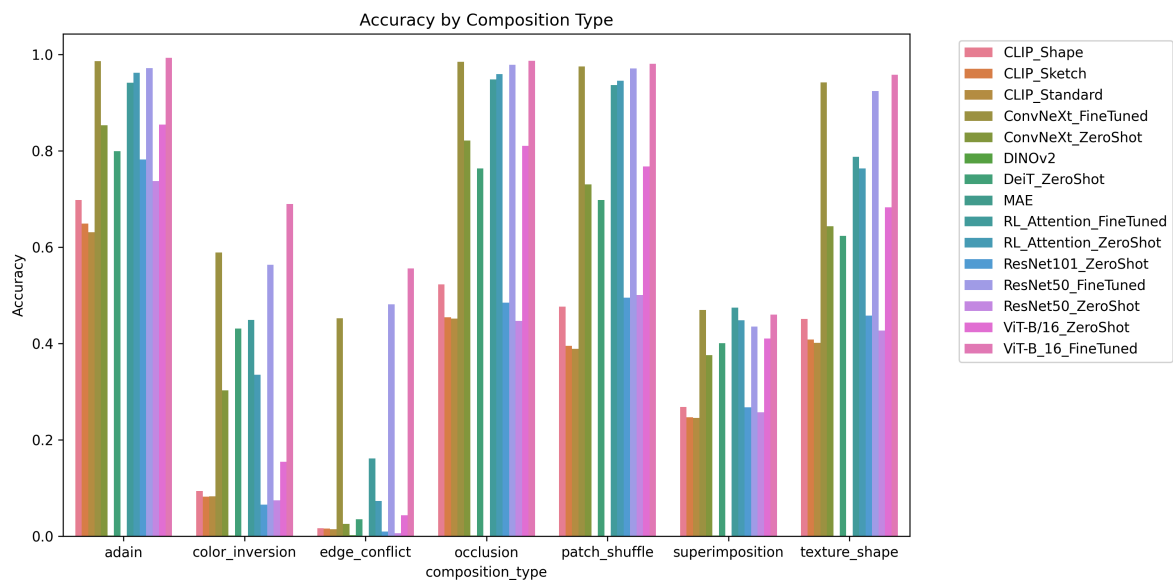


Figure 4: Breakdown of accuracy metrics grouped by composition type.

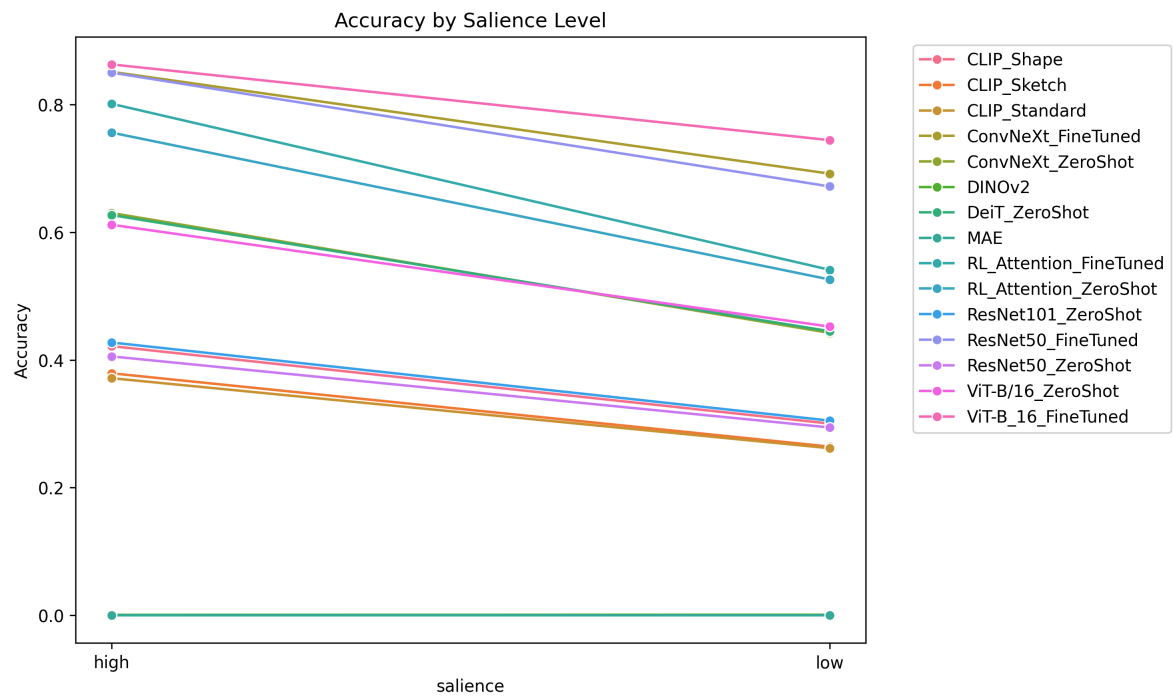


Figure 5: Performance degradation when moving from high to low shape saliency.

## 4. Error Consistency

Measuring prediction agreement between model pairs to identify shared decision boundaries.

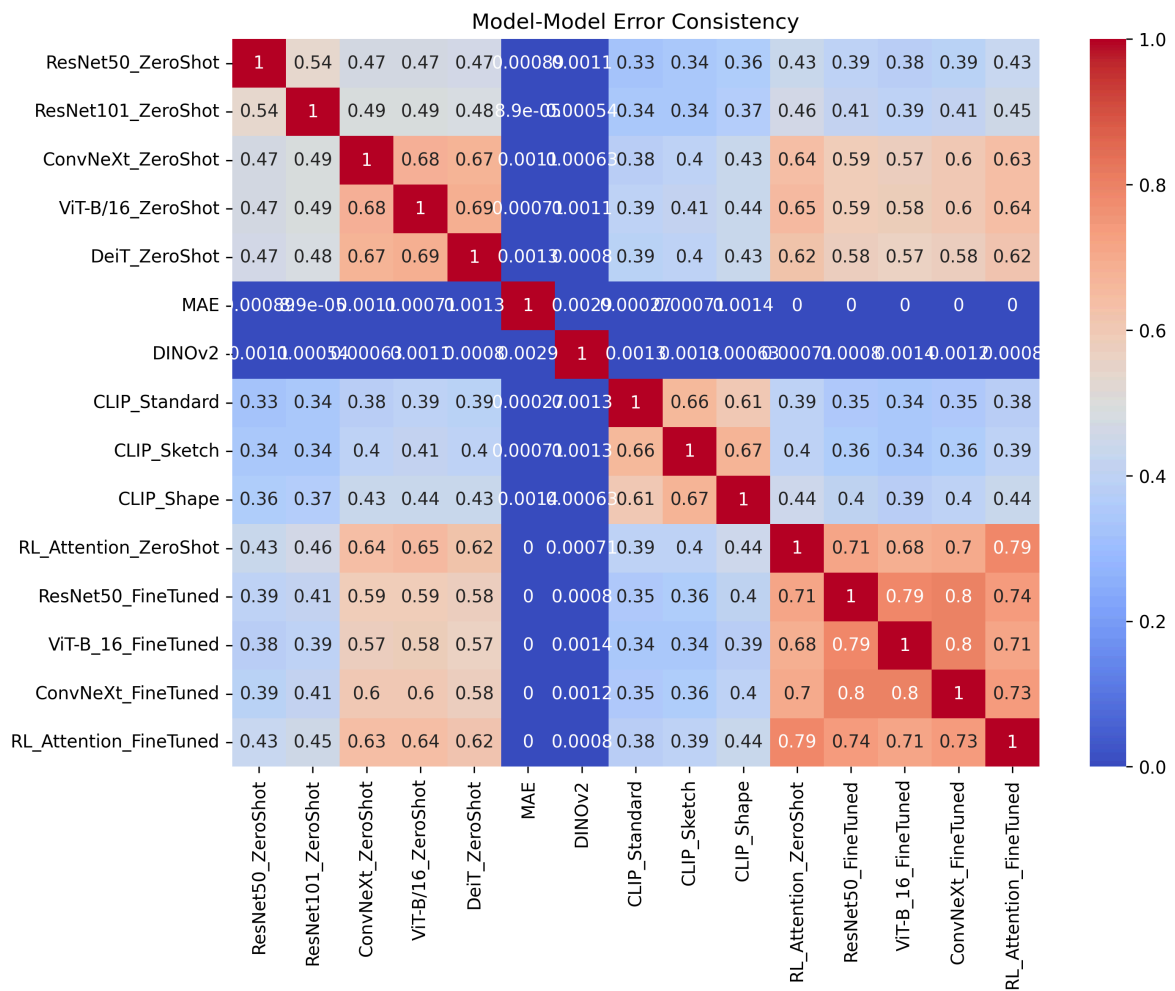


Figure 6: Model-Model error consistency. Fine-tuned models form a distinct cluster of high agreement.

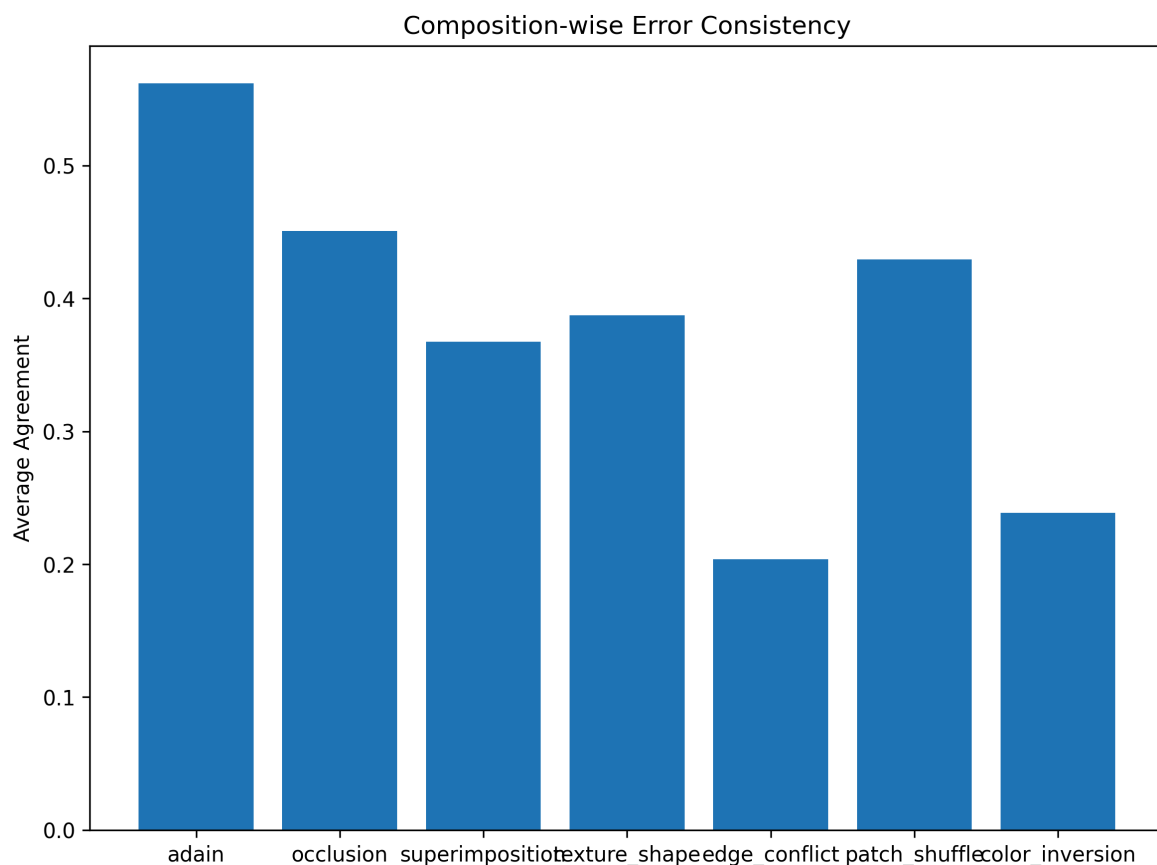


Figure 7: Consistency clustered by composition method.

## 5. Statistical Significance Highlights

Based on paired McNemar's tests across 105 pairwise comparisons, 97 differences remain statistically significant after Bonferroni correction ( $\alpha = 0.000476$ ).

- **Zero-Shot Advantage:** `RL_Attention_ZeroShot` vs. `ResNet50_ZeroShot` (+29.10pp) is statistically significant ( $p = 0.0$ ), with a medium effect size (Cohen's  $h = 0.59$ ).
- **Fine-Tuning Impact:** `ResNet50_FineTuned` vs. `ResNet50_ZeroShot` (+41.08pp) is statistically significant ( $p = 0.0$ ), with a large effect size (Cohen's  $h = 0.85$ ).
- **Indistinguishable Pairs:** `ConvNeXt_ZeroShot` vs. `DeiT_ZeroShot` (+0.03pp) is **not** statistically significant ( $p = 0.960$ ).

## 6. Inference Profiling

The computational cost of models during inference.

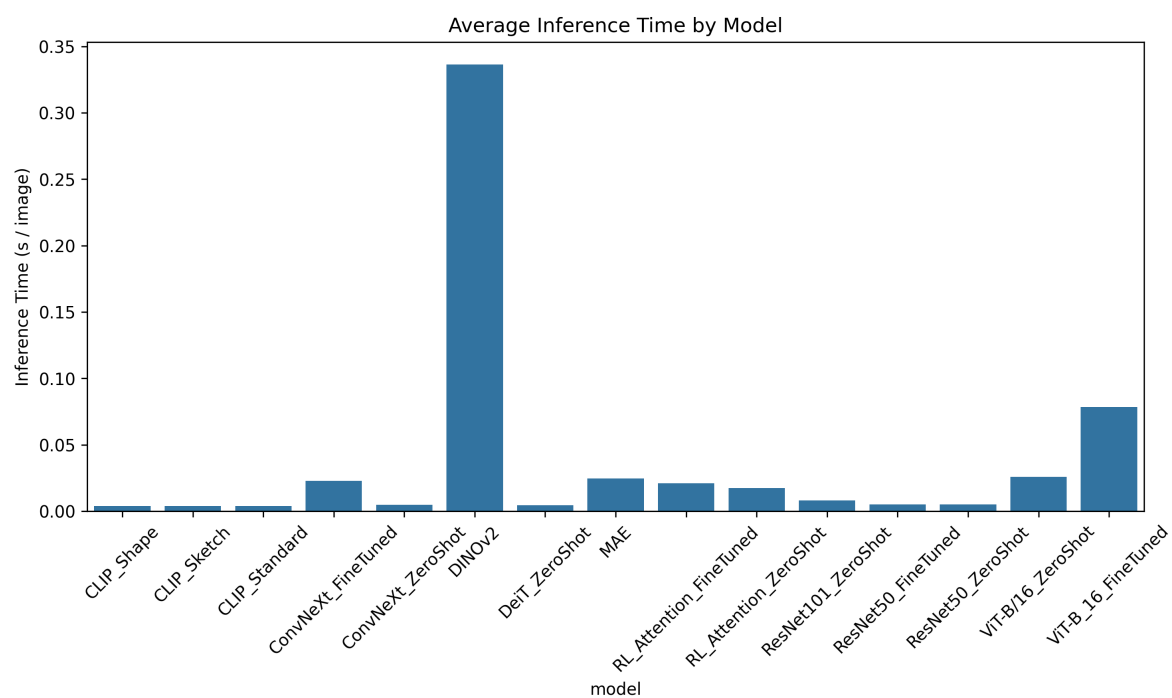


Figure 8: Average inference time (in seconds).